

$$\textcircled{1} \underline{dw} \sim N(0, dt)$$

$$= (\sqrt{dt}) \cdot Z \quad \text{where } Z \sim N(0, 1)$$

$$Z^2 \sim \chi^2_{(1)} \\ \swarrow \quad \searrow \\ \mathbb{E} = 1 \quad \mathbb{V} = 2$$

$$\mathbb{E}[dw] = 0, \quad \mathbb{V}(dw) = dt$$

$$\begin{cases} \mathbb{E}[dw^2] = \mathbb{E}[dt \cdot Z^2] = (dt) \mathbb{E}[Z^2] = dt \\ \mathbb{V}[dw^2] = \mathbb{V}[dt \cdot Z^2] = (dt)^2 \times \mathbb{V}(Z^2) = 2(dt)^2 \rightarrow 0 \end{cases}$$

$$\underline{dw^2} \Rightarrow (dt, 0) \text{ -- (deterministic)}$$

$$\mathbb{E}[dw \cdot dt] = dt \cdot \mathbb{E}[dw] = 0$$

$$(dt)^{(>1)} \rightarrow 0$$

$$\mathbb{V}[dw \cdot dt] = (dt)^2 \times dt = (dt)^3 \rightarrow 0$$

$$\lim_{x \rightarrow 0} \left(\frac{2x^2}{5x^2} \right) = \left(\frac{2}{5} \right)$$

$$\lim_{x \rightarrow 0} \frac{12x^2}{5x^2} \leftarrow$$

$$\frac{|dy|}{|dx|} \quad \frac{dy}{dx} \\ \begin{matrix} \nearrow & \searrow \\ 0 & \infty \\ dy \ll dx & dy \gg dx \end{matrix}$$

$$\begin{matrix} dt^2 & \rightarrow 0 \\ dt^{3/2} & \rightarrow 0 \\ dt^3 & \rightarrow 0 \end{matrix}$$

$$\textcircled{dt}$$

$$W_t \sim N(0, t)$$

$$= z\sqrt{t}$$

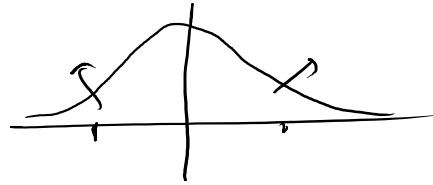
$$\mathbb{E}[W_t] = 0, \quad \forall [W_t] = t$$

$$\mathbb{E}[W_t^2] = \mathbb{E}[t \cdot z^2] = t$$

$$\mathbb{V}[W_t] = \mathbb{V}[t \cdot z^2] = t^2$$

$$\mathbb{E}[W_t^3] = \mathbb{E}[t^{3/2} \cdot z^3] = t^{3/2} \cdot \mathbb{E}[z^3] = 0$$

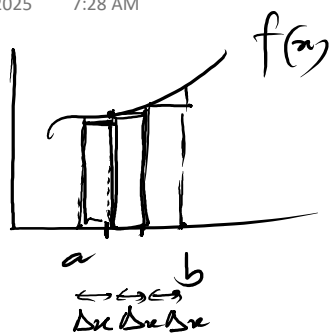
$$\mathbb{E}[W_t^4] = \mathbb{E}[t^2 \cdot z^4] = 3t^2$$



$$\mathbb{E}[z^1] = 0$$

(skewness $\Rightarrow$)

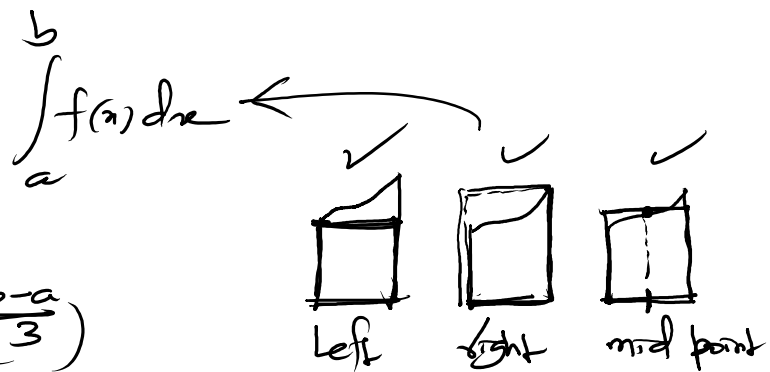
$$\mathbb{E}[z^k] = 0 \quad k \text{ is odd}$$



$$n = 3$$

$$\Delta x = \left(\frac{b-a}{3} \right)$$

$$n \rightarrow 6$$



$$\lim_{n \rightarrow \infty} \sum_{i=1}^n W_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) \quad \leftarrow \int_0^t W_s dW_s$$

$$\leq (W_{t_{i-1}} \cdot W_{t_i} - W_{t_{i-1}}^2)$$

dW

$$\leq \frac{1}{2} (2 W_{t_{i-1}} \cdot W_{t_i} - 2 W_{t_{i-1}}^2) \rightarrow$$

$$\leq \frac{1}{2} \left(- (W_{t_i} - W_{t_{i-1}})^2 + W_{t_i}^2 - W_{t_{i-1}}^2 \right)$$

$$= -\frac{1}{2} \Delta W^2 + \frac{1}{2} (W_{t_i}^2 - W_{t_{i-1}}^2)$$

$$= -t/2 + \frac{1}{2} (W_t^2)$$

$$= \frac{1}{2} (W_t^2 - t)$$

$$\begin{aligned} & W_1^2 - W_0^2 \\ & + W_2^2 - W_1^2 \\ & + \cancel{W_3^2 - W_2^2} \\ & + \cancel{W_4^2 - W_3^2} \\ & + \dots \\ & + W_n^2 - W_{n-1}^2 \end{aligned}$$

$$I_t = \int_0^t f(s, W_s) dW_s$$

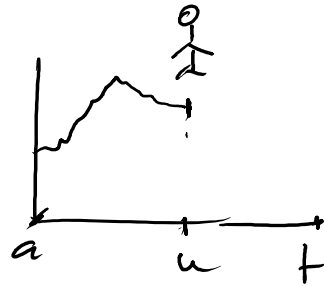
$$\begin{aligned} \mathbb{E}[I_t] &= \mathbb{E}\left[\sum f(s, W_s) \cdot (\Delta W_s)\right] \\ &= \sum \mathbb{E}\left[f(s, W_s) \cdot (\Delta W_s)\right] \\ &= \sum \mathbb{E}\left[f(s, W_s)\right] \mathbb{E}\left[(\Delta W_s)\right] \\ &= 0 \end{aligned}$$

$$X_t = \int_a^t f(s, W_s) dW_s \quad \text{--- (ito integral)}$$

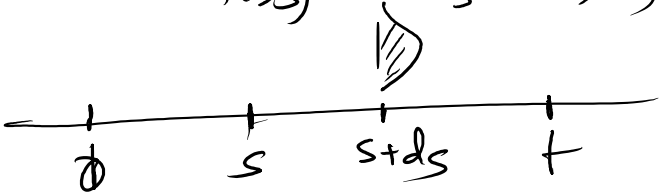
$$\rightarrow \mathbb{E}[X_t | \mathcal{F}_u] = X_u \quad \rightarrow \text{martingale}$$

$$= \mathbb{E}\left[\underbrace{\int_a^u f(s, W_s) dW_s} + \int_u^t f(s, W_s) dW_s\right]$$

$$= \underbrace{\int_a^u f(s, W_s) dW_s} + 0 = X_u$$

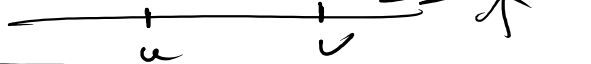


$f(s, w_s)$ $dw_s \sim N(0, ds)$



$\mathbb{E} \left[\left(\int_0^t f(s, w_s) dw_s \right)^2 \right]$

$f(u, w_u) dw_u$ $f(v, w_v) dw_v$



$\int_0^t f(s, w_s) ds$

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